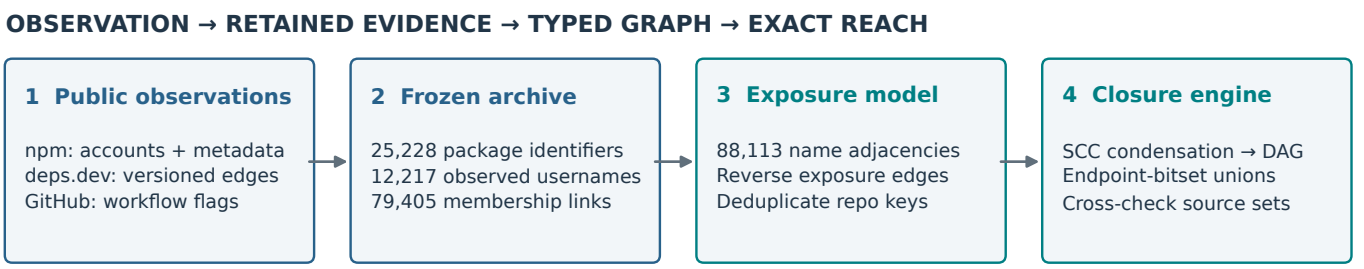
OBSERVATION → RETAINED EVIDENCE → TYPED GRAPH → EXACT REACH
1 Public observations
npm: accounts + metadata
deps.dev: versioned edges
GitHub: workflow flags
2 Frozen archive
25,228 package identifiers
12,217 observed usernames
79,405 membership links
3 Exposure model
88,113 name adjacencies
Reverse exposure edges
Deduplicate repo keys
4 Closure engine
SCC condensation → DAG
Endpoint-bitset unions
Cross-check source sets
Typed relations:
maintainer → package → dependent → repository
membership
reverse reach
lexical mapping
7,171 workflow repositories • 7,158 with observed sources • directory presence ≠ executed CI